

CGI Capacity Analyzer for Director Workload and RFP Assignment

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Abstract

CGI Atlantic / Media Atlantic needs a more consistent, data-informed way to understand director workload and assign new RFPs. We develop a Capacity Analyzer that combines historical CRM opportunity data with proposal and RFP text to support director capacity estimation and RFP assignment. The system is designed as a transparent decision-support tool rather than a replacement for business judgment. Week 1 established the reproducible opportunity data foundation and module skeletons. Week 2 extended this foundation into a first integrated prototype by refining the cleaned opportunity dataset, validating scoring fields, implementing director-level capacity scoring, connecting capacity outputs to the Streamlit dashboard, and advancing the RFP preprocessing pipeline. Week 3 refined the capacity dashboard around a relative-load-first interpretation and shifted the main focus toward RFP assignment, including fallback retrieval, assignment schema stabilization, heuristic recommendation logic, and Streamlit RFP page integration. Week 4 expanded the prototype into a final stakeholder-facing version with real capacity-aware RFP recommendations, Azure OpenAI chat enrichment for RFP summaries, effort estimates, and director match reasons, Azure OpenAI embedding and Chroma retrieval integration with local fallback preserved, RFP upload support alongside paste input, clearer fallback and prototype-status labeling, and stronger empty-state, error-state, and response-quality checks. Week 5 focused on final app QA and refinement rather than major new feature development: the team validated Azure/Chroma retrieval and fallback behavior, reviewed whether retrieved examples supported displayed RFP outputs, refined assignment caveats and risk wording, improved stakeholder-facing labels and LLM/retrieval status captions, and consolidated final scope, test, and caveat tracking. The resulting prototype remains a transparent stakeholder-facing decision-support sys-

tem rather than a production-ready or business-validated assignment system.

1 Introduction

CGI Atlantic / Media Atlantic needs a more consistent, data-informed way to understand director workload and assign new RFPs. Today, capacity may be tracked manually or inconsistently, making it difficult to see which directors are available, near their historical workload norm, overloaded, or well matched to a new opportunity.

This project builds a CGI Capacity Analyzer to support two related business decisions. First, the Director Capacity Dashboard summarizes each director's current workload, historical baseline, relative load, capacity label, and workload drivers. In the Week 5 refined prototype, `relative_load` remains the primary business-facing metric because it compares a director's current workload against that same director's historical norm. Second, the RFP Assignment Tool estimates RFP effort, retrieves semantically similar historical proposal or response examples through the preferred Azure OpenAI embedding and Chroma retrieval path when available, and recommends suitable directors based on available capacity signals, retrieved supporting examples, service-domain overlap, and risk. The local fallback retrieval path remains available when Azure or Chroma is unavailable, skipped, or unsuitable for the current request, and the RFP page reports the active retrieval mode so stakeholders can distinguish Azure/Chroma-backed retrieval from local fallback retrieval.

The system is not intended to replace stakeholder judgment. Instead, it provides a transparent and reproducible data-based starting point that CGI stakeholders can review, validate, and adjust. This design choice is important because the project does not currently have manually labeled ground truth for director capacity, RFP effort, or correct director assignment. The Week 5 RFP outputs should there-

fore be interpreted as prototype decision-support guidance: capacity-aware when real or pre-built capacity data is available, LLM-enriched when Azure chat configuration is active, and still supported by deterministic heuristic or fallback behavior when LLM enrichment or retrieval components are unavailable. Week 5 user-style QA and evidence review further clarified that retrieved examples should be treated as semantic evidence related to the submitted RFP text, not as proof of prior director involvement, confirmed director experience, or CRM opportunity ownership. Recommendations remain stakeholder-facing prototype outputs that require CGI review and validation before any staffing decision.

2 Data

2.1 Data Sources

The project uses two anonymized Excel CRM opportunity exports and one JSON file containing historical proposal and response text. The two opportunity exports are referred to as opps1 and opps2. The proposal data contains historical proposal and response document pairs.

All CGI-provided files are treated as sensitive local data. They are not committed to GitHub. Local credentials and API configuration are also excluded from version control.

2.2 Opportunity Data

Initial EDA indicates that opps1 and opps2 are different views of largely overlapping CRM opportunity data. We use opps2 as the primary base table because it contains the richer CRM schema and has one unique opportunity_id per row in the current export. We use opps1 as a secondary source: selected fields from opps1 are collapsed to one row per opportunity_id and merged into opps2, while opps1-exclusive opportunities are appended to the final opportunity_df. Fields that appear under different names across the two exports, especially revenue-related fields, require normalization and validation before downstream scoring and are handled through the Week 2 validation and fallback process.

This distinction is important because the downstream capacity model is expected to operate at the opportunity level. During the Week 1 merge implementation, we found that opps1 contains duplicate opportunity_id values. These duplicates likely reflect service-level or product-level detail

rather than distinct opportunities. A direct merge from opps1 into opps2 would therefore multiply some opportunity rows and risk double-counting workload in later capacity scoring.

To preserve the intended data grain, the merge pipeline collapses opps1 supplemental fields to one row per opportunity_id before joining them to opps2. Duplicate and detail rows from opps1 are preserved in local audit outputs rather than being used to expand the final opportunity_df.

2.3 Proposal and RFP Text Data

The proposal JSON contains 19 historical proposal and response document pairs. These documents are text-heavy and many are too long to embed as single inputs. Therefore, the RFP pipeline prepares proposal text through text extraction, chunking, metadata construction, and retrieval-oriented processing. The historical or sample corpus is chunked before retrieval so that retrieved evidence can be surfaced as smaller supporting examples rather than as full proposal documents.

By Week 5, the RFP pipeline loads and chunks the local proposals_responses.json corpus when available, with a built-in sample corpus used only as fallback. Submitted RFP text, whether pasted or extracted from TXT, DOCX, or text-based PDF upload, is used as query text against the chunked historical or sample corpus. The preferred retrieval path uses Azure OpenAI embeddings and Chroma when configured and suitable for the request, while deterministic local retrieval over the proposal corpus remains available when Azure / Chroma is unavailable, skipped, or fails. The pipeline then generates a dashboard-facing assignment_context for Streamlit display.

The proposal text supports the RFP Assignment Tool by enabling retrieval of semantically related historical or sample examples. For a new RFP, the system retrieves related proposal or response chunks and combines retrieval evidence with structured capacity data, service-domain overlap, and risk flags. In Week 4, the RFP page was updated to pass real or pre-built capacity_df into generate_assignment_context() when available, so recommended directors can be capacity-aware rather than purely mock-backed. Week 5 QA reconfirmed this handoff and refined the output wording so capacity-aware recommendations are still presented as decision-support guidance rather than final staffing decisions. If only synthetic capacity data is available, the dashboard labels this

Data source	Approx. size	Main contents	Role in project
opps1 CRM export	8,435 rows 20 columns	Supplemental product, service-solution, IP, service-solution revenue, opportunity-level fallback revenue, and delivery territory fields; contains repeated opportunity_id values.	Secondary supplemental source. Selected fields are collapsed to one row per opportunity_id before merging into opps2.
opps2 CRM export	8,695 rows 33 columns	Richer CRM opportunity schema with owner, status, sales stage, probability, revenue, date, duration, and proposal-related fields.	Primary base table for opportunity_df. It defines the opportunity-level grain for capacity scoring and dashboard inputs.
Proposal / response JSON	19 document pairs	Historical proposal and response text.	Used for RFP text preprocessing, historical corpus chunking, retrieved supporting examples, assignment-context generation, Azure/Chroma backend retrieval validation, and fallback-safe local retrieval.

Table 1: Overview of the main project data sources used through Week 5. The files are stored locally and are not committed to GitHub because they contain CGI-provided data.

clearly and treats the output as fallback prototype guidance.

After CGI provided the remaining Azure configuration values, Azure embedding access was validated with a real smoke test, and the Azure embedding plus Chroma retrieval path was integrated into the active RFP backend. In Week 5, retrieval and configuration validation further confirmed that Azure configuration is readable locally, Azure embedding calls produce vectors, Azure/Chroma retrieval works on both sample and full proposal-corpus backend checks, and `generate_assignment_context()` can return `retrieval_mode` and `retrieval_status` metadata. The Streamlit RFP page reads this backend metadata and displays the active retrieval mode, allowing stakeholders to distinguish Azure/Chroma retrieval from local fallback retrieval. Unless an explicit browser-based Azure/Chroma app-flow run is documented, these Azure/Chroma results should be described as backend smoke-test, sample, and full-corpus retrieval validation rather than full production app validation.

A key remaining data limitation is that retrieved proposal chunks cannot currently be tied reliably to specific directors or CRM opportunities. Week 5 evidence review and proposal-to-CRM crosswalk checks confirmed that most proposal/RFP entries do not have strong automatic CRM matches. Therefore, retrieved examples should be interpreted as semantic evidence related to the submitted RFP text, not as proof that a recommended director per-

sonally worked on a retrieved historical proposal or owned a matched CRM opportunity.

2.4 EDA Findings and Design Implications

The EDA directly influenced several early design decisions. First, because opps2 has the richer CRM schema and preserves the opportunity-level grain more cleanly, it is used as the base table for `opportunity_df`. Second, because opps1 contains useful service and product fields but also duplicated opportunity IDs, it is treated as a supplemental source rather than a second base table. Third, because the proposal documents are long, the RFP pipeline uses chunking for the historical or sample corpus before embedding or retrieval-style processing. Finally, because the data does not contain labeled capacity categories or gold-standard RFP assignments, the system uses transparent formula-based scoring and retrieval-supported recommendation logic instead of supervised learning.

By Week 5, these design implications became more concrete for the refined final prototype. The capacity dashboard continues to rely on opportunity-level CRM data and formula-based workload estimation, while the RFP Assignment Tool combines Azure/Chroma or fallback retrieval evidence with structured capacity signals when `capacity_df` is available. The team validated Azure embedding access, confirmed Azure/Chroma backend retrieval over both sample and full proposal-corpus checks, preserved local fallback retrieval for robustness, and refined the

Streamlit wording so retrieval mode, LLM enrichment status, capacity-data source, and prototype caveats are clearer to stakeholders.

The final prototype therefore distinguishes between the preferred Azure/Chroma retrieval path, local fallback retrieval behavior, heuristic scoring, LLM-enriched explanation fields, semantic retrieval evidence, and remaining business-validation limitations. Because reliable proposal-to-director and proposal-to-opportunity linkage is still unavailable, retrieved supporting examples can help stakeholders inspect semantic similarity, but they should not be overinterpreted as confirmed evidence of director experience or final assignment suitability.

3 System Overview

The project has two core deliverables. The first is a Director Capacity Dashboard. This dashboard uses historical CRM opportunity data to estimate each director's current workload, compare the current workload against the director's historical baseline, and display capacity labels such as Available, Near Historical Norm, High Load, Overextended, and No baseline. In the Week 5 refined prototype, `relative_load` remains the primary business-facing workload metric because it directly compares a director's current workload against that same director's historical norm.

The dashboard also exposes drivers of the score, including open opportunities, late-stage opportunities, weighted pipeline, and inferred delivery commitments. This is important because stakeholders need not only a final capacity label, but also an explanation of why a director appears available, near their normal workload, highly loaded, or overextended. The dashboard keeps `capacity_score` in the backend for compatibility, but the main dashboard interpretation is now based on `relative_load`.

The second deliverable is an RFP Assignment Tool. For a new RFP, the tool estimates effort, retrieves supporting proposal or response examples from the historical or fallback corpus, and recommends directors using retrieval evidence, available capacity signals, service-domain overlap, and risk flags. By Week 4, the RFP page was updated to pass real or pre-built `capacity_df` into `generate_assignment_context()` when available, so the recommended directors can be capacity-aware rather than purely mock-backed. Week 5 QA reconfirmed this capacity handoff and refined the

stakeholder-facing wording so recommended directors are presented as prototype decision-support suggestions rather than final staffing decisions. The RFP output also supports Azure OpenAI chat enrichment for the RFP summary, effort estimate, and director match reasons. If chat enrichment is unavailable, fails, or returns malformed output, the system falls back to deterministic heuristic output with the same dashboard-facing schema.

The current architecture is modular. The opportunity cleaning pipeline produces a local `opportunity_df` and a cleaned downstream dataset for validation and scoring. The capacity scoring module consumes this data to compute director-level workload, `relative_load`, `capacity_score`, and capacity labels. The Streamlit dashboard displays capacity metrics and provides an interface for RFP assignment. The RFP pipeline preprocesses proposal and response text, creates historical or sample corpus chunks for retrieval, supports the preferred Azure/Chroma backend retrieval path when available, preserves local fallback retrieval, and produces a dashboard-facing `assignment_context` schema for downstream recommendation display.

After CGI provided the remaining Azure configuration values in Week 4, Azure embedding access was validated through a real smoke test, and Azure-generated embeddings were successfully added to and queried from Chroma. In Week 5, retrieval and configuration validation further confirmed that Azure/Chroma retrieval works at backend smoke-test, sample, and full-corpus levels, including retrieval over the local proposal corpus and `retrieval_mode / retrieval_status` metadata returned through `generate_assignment_context()`. The local fallback retrieval path remains available if Azure embedding, Chroma storage, Chroma querying, or vector-store setup is unavailable. The RFP page reads and displays the active retrieval mode from the backend output so stakeholders can distinguish Azure/Chroma retrieval from local fallback retrieval.

Therefore, the final prototype should be interpreted as a transparent stakeholder-facing decision-support prototype with capacity-aware recommendations when capacity data is available, LLM-enriched output when configured, Azure/Chroma retrieval when the preferred backend path is active, fallback-safe behavior, and clearly documented limitations. Week 5 evidence review and usability re-

Merge statistic	Value
opps1 rows	8,435
opps2 rows	8,695
opps1 unique IDs	7,679
opps2 unique IDs	8,695
Matched opportunity IDs	7,628
opps1-exclusive IDs	51
opps2-only IDs	1,067
Final opportunity_df rows	8,746
Duplicate IDs in final output	0

Table 2: Week 1 opportunity merge audit summary. The final output preserves opportunity-level grain by avoiding one-to-many expansion from duplicated opps1 records.

view further clarified that retrieved examples are semantic supporting evidence, not proof of prior director involvement or confirmed CRM opportunity ownership. The prototype still requires CGI validation before any staffing or assignment decisions.

4 Weekly Implementation Progress

4.1 Week 1: Data Foundation and Module Skeletons

During Week 1, the team focused on building the first reproducible data foundation and creating skeletons for the main system components. The goal was not to finalize the full capacity model, dashboard, or RFP assignment tool. Instead, we prioritized stable data structures, module boundaries, and interfaces that can support later development.

The work was divided across five roles. Yixiao focused on opportunity data merging, schema comparison, and duplicate / unmatched record summaries. Kian focused on validation of revenue, date, duration, probability, status, and sales-stage fields. Lyken focused on the capacity scoring skeleton and director table prototype. Freya focused on the Streamlit dashboard skeleton and RFP page placeholder. Jai focused on the RFP preprocessing and vector store skeleton.

For the opportunity data merge, we implemented a reproducible pipeline that loads the two anonymized CRM Excel exports, normalizes column names, compares schemas, uses opps2 as the base opportunity-level table, and supplements selected fields from opps1. The pipeline uses opportunity_id as the required join key.

A key Week 1 finding was that opps1 contains duplicate opportunity_id rows. These duplicates likely reflect service-level or product-level detail rather than separate opportunities. A direct merge

from opps1 into opps2 created one-to-many expansion, which would risk double-counting workload in downstream capacity scoring.

To preserve the intended opportunity-level grain, we collapsed opps1 supplemental fields to one row per opportunity_id before merging. Duplicate and detail rows from opps1 are preserved in local audit outputs instead of multiplying the final opportunity_df. The final local opportunity_df contains 8,746 rows, matching the expected no-expansion count of 8,695 opps2 base rows plus 51 opps1-exclusive opportunity IDs. The final merged dataset contains zero duplicate opportunity_id rows.

All generated processed data files are kept local and are not committed to GitHub because they are derived from sensitive CGI data. The reproducible code and tests are committed instead, so other team members with local access to the source files can regenerate the same outputs.

4.2 Week 2: From Skeletons to First Integrated Prototype

In Week 2, the team moved from separate data and module skeletons toward a first integrated prototype. The main development path connected the cleaned opportunity data to validation, owner-level capacity scoring, and Streamlit dashboard display. In parallel, the RFP pipeline progressed from placeholder functions toward sample text preprocessing, chunking, and retrieval-style output.

Yixiao refined the Week 1 merge output into a cleaner downstream dataset for validation, scoring, and dashboard development. This work focused on stabilizing cleaned_opportunity_df, confirming the required columns for downstream modules, preserving source and quality flags, and preparing an owner-level base summary using opportunity_owner. The goal was not to redesign the Week 1 merge logic, but to make the merged data easier for Kian's validation work, Lyken's scoring module, and Freya's dashboard integration to consume.

Kian extended the validation work from basic field inspection to scoring-oriented field assessment. The validation process reviewed revenue fields, date and duration fields, probability, status, sales stage, opportunity_owner, and opportunity_manager. The output separates fields that are reliable, fields that are usable with fallback assumptions, and fields that remain risky for formal scoring. This validation layer is impor-

Module	Owner	Week 2 update	Main output
Data integration	Yixiao	Refined the Week 1 merge output into a downstream-ready cleaned dataset.	cleaned_opportunity_df, column dictionary, quality flags, and owner-level summary.
Data validation	Kian	Validated fields needed for capacity scoring and dashboard use.	Missingness summary, reliable / risky fields, revenue fallback hierarchy, and date / duration assumptions.
Capacity scoring	Lyken	Implemented the first working owner-level capacity scoring workflow.	director_capacity_df, relative load, capacity score, capacity label, and scoring sanity checks.
Dashboard	Freya	Connected the dashboard to real or schema-aligned scoring output.	Capacity score chart, owner-level table, filters, and explanation placeholders.
RFP pipeline	Jai	Built an independently runnable RFP preprocessing and retrieval-style prototype.	Sample chunks, mock or basic retrieval output, and assignment_context schema.
Consistency pass	Yixiao, Freya	Aligned architecture, capacity scoring, dashboard behavior, RFP retrieval, and output contracts after the latest RFP/capacity merge.	Fixed scoring gates, active-quarter baseline/trend logic, dashboard live-data path, stale RFP retrieval, and related docs/tests.

Table 3: Week 2 progress from separate skeletons toward the first integrated prototype.

tant because the capacity model depends on transparent assumptions rather than supervised labels.

Lyken converted the Week 1 `capacity_engine.py` skeleton into a first working capacity scoring workflow. The scoring module computes sales load, inferred delivery load, current owner-level load, historical baseline, relative load, bounded capacity score, and capacity labels. The resulting director-level table provides the first version of `director_capacity_df`, including fields such as `current_load`, `historical_avg_load`, `relative_load`, `capacity_score`, and `capacity_label`.

Freya updated the Streamlit dashboard from a mock-data skeleton toward an integrated capacity dashboard. The Week 2 dashboard reads either the first `director_capacity_df` output or a schema-aligned sample output with the same expected columns. The dashboard displays capacity score by director, capacity labels, current load versus historical baseline, and an owner-level capacity table. Filters and explanation placeholders were also retained so that the interface can continue to evolve as scoring assumptions are refined.

Jai advanced the RFP pipeline from skeleton files toward an independently runnable prototype. The Week 2 work focused on RFP text preprocessing, chunk generation, chunk metadata structure, and a stable `assignment_context` output schema for dashboard integration. The prototype can produce sample chunks and basic or mock retrieval-style output. Full Azure OpenAI and Chroma integra-

tion remain later steps, but the Week 2 interface gives the dashboard a stable structure for retrieved examples, recommended directors, risk flags, and explanatory notes.

Overall, Week 2 established the first end-to-end project workflow: cleaned opportunity data supports validation; validation findings inform capacity scoring; capacity scoring produces a dashboard-ready director table; and the dashboard begins displaying capacity outputs. The RFP pipeline remains partly independent from the capacity workflow, but it now has a clearer preprocessing and output contract for future recommendation integration.

4.3 Week 3: Capacity Dashboard Refinement and RFP Assignment Integration

In Week 3, the team shifted the main development focus from the opportunity data foundation and capacity dashboard prototype toward the RFP Assignment Tool. At the same time, the Director Capacity Dashboard received an important refinement: the dashboard now treats `relative_load` as the primary business-facing workload metric instead of `capacity_score`. This change better communicates overextension because `relative_load` compares a director's current workload against that same director's historical norm, while `capacity_score` is capped once a director reaches or exceeds baseline.

The Week 3 capacity dashboard update also clarified the workload formula and label interpretation.

Current load is represented as the sum of sales load and delivery load. Sales load uses open opportunities, stage weight, probability, authoritative revenue, and duration weight. Delivery load uses won active deliveries and monthly revenue. Capacity labels are now assigned from `relative_load`: *Available*, *Near Historical Norm*, *High Load*, *Overextended*, and *No baseline*. These thresholds are project-side prototype choices and should be reviewed by CGI before production use.

The dashboard filtering logic was also refined. Territory and opportunity-owner filters define the selected director population for both current workload and historical baseline. Row-level filters such as status, sales stage, opportunity type, sales model, and created-date range narrow the current workload and trend views but do not recompute the historical baseline denominator. This keeps the baseline stable and makes filtered views answer a clearer business question: how the filtered current workload compares with the selected directors' normal workload.

In parallel, the RFP pipeline became the main Week 3 development focus. Based on repository review, the local RFP prototype is partially ready: it can process available proposal text, create chunks, run local fallback embedding or in-memory retrieval, and generate a dashboard-facing `assignment_context`. The active demo path is fallback / heuristic rather than Azure-backed. The remaining integration work focuses on confirming Azure API configuration, completing real Azure OpenAI embedding smoke tests, connecting Chroma retrieval end to end, wiring real capacity data into the RFP engine, refining recommendation logic, and continuing Streamlit RFP page display work.

Yixiao coordinated the Week 3 RFP integration and documentation work. This included reviewing which RFP pipeline components currently work locally, confirming the intended API and fallback retrieval behavior, aligning the `assignment_context` schema with Jai and Freya, and documenting reproducibility, assumptions, blockers, and risks. A key schema decision is that the backend can keep the current field name `effort`, while the dashboard displays this object as Estimated Effort. If time allows, `estimated_effort` can be added later as an alias without disrupting existing tests.

Kian completed Sprint 3 implementing a reproducible RFP validation workflow over

`proposals_responses.json`. The validation confirmed 19 top-level records and 46 extracted documents (19 proposal documents and 27 response documents), with no malformed top-level records in the current local dataset. The work formalized a chunk metadata contract and validated required fields (`proposal_id`, `chunk_id`, `document_id`, `source_type`, `chunk_index`, `text`, `opportunity_owner`, `opportunity_id`) across 1,216 generated chunks, with zero missing required fields. A key finding is that proposal chunks currently cannot be reliably linked to directors or opportunities because `opportunity_owner` and `opportunity_id` are null in all validated chunks. This limitation is documented explicitly rather than patched with unreliable inferred mapping. Deliverables include `src/rfp_data_validator.py`, `scripts/validate_rfp_data_and_export_samples.py`, `docs/rfp_data_validation.md`, and exported artifacts under `outputs/rfp_validation/` for Lyken/Jai handoff. Export defaults use redacted text and anonymized IDs for safe sharing.

Lyken focused on the Azure embedding and vector-store path. The Week 3 goal was to move beyond local heuristic embedding toward Azure OpenAI embeddings and Chroma retrieval, while keeping local fallback retrieval available. This work included confirming environment-variable requirements, implementing or updating the Azure client, connecting embeddings to `vector_store.py`, and documenting how the system should behave when Azure is unavailable. At the end of Week 3, real Azure embedding smoke testing was still blocked because the required API version and embedding deployment name had not yet been confirmed, so fallback retrieval remained the stable path for RFP dashboard development.

Freya focused on the RFP page in Streamlit. The Week 3 target was to move the RFP Assignment Tool page beyond a placeholder by displaying `assignment_context` fields such as RFP summary, estimated effort, retrieved examples, recommended directors, capacity signals, match reasons, supporting chunks, and risk flags. The page also needed to handle empty retrieval results or API failures gracefully and clearly label prototype, heuristic, fallback, or mock outputs.

Jai focused on improving `generate_assignment_context()` and the recommendation logic. The Week 3 output used retrieved examples as supporting evidence and could incorporate capacity fields when director

Track	Owner	Week 3 focus	Main output
Capacity dashboard refinement	Freya	Implemented and documented the relative_load-first dashboard interpretation and updated capacity-label semantics.	Updated dashboard charts, filter semantics, workload formulas, documentation, and tests.
RFP integration / QA	Yixiao	Coordinated RFP pipeline status, API / fallback behavior, assignment schema, smoke testing, and documentation.	API usage notes, finalized assignment_context schema, smoke test checklist, integration QA notes, and risks.
RFP data validation	Kian	Validated proposals_responses.json, text extraction quality, chunk metadata, and proposal-to-director linkage limitations.	JSON structure summary, chunk metadata dictionary, sample chunks, and RFP data validation notes.
Azure embedding / vector store	Lyken	Advanced the code path from local fallback toward Azure embeddings and Chroma while preserving fallback retrieval; real Azure smoke testing remained blocked by missing API version and deployment configuration.	Azure client helper updates, vector store structure, fallback retrieval behavior, and documented blockers for Azure / Chroma completion.
RFP dashboard integration	Freya	Moved the Streamlit RFP page from placeholder toward displaying assignment_context.	RFP input UI, result display, retrieved examples, recommended directors, risk flags, and display notes.
RFP assignment logic	Jai	Improved generate_assignment_context() toward heuristic recommendation output using retrieval examples, available capacity fields when supplied, and risk flags.	Stable assignment output, heuristic effort logic, recommendation notes, risk flag logic, and test updates.

Table 4: Week 3 progress toward RFP Assignment Tool integration.

data was supplied. However, the Streamlit RFP page had not yet been updated to pass the loaded capacity data into the engine, so the Week 3 recommendations were still best described as heuristic or mock-backed prototype output rather than fully capacity-aware assignment guidance.

Overall, Week 3 moved the project from a capacity-dashboard-centered prototype toward a broader integrated decision-support tool. The dashboard gained a clearer relative-load-first business interpretation, while the RFP Assignment Tool gained a more explicit path from local fallback retrieval toward Azure embedding, Chroma retrieval, assignment schema stabilization, recommendation logic, and Streamlit display. These Week 3 limitations directly shaped the Week 4 priorities: final integration QA, real capacity-data handoff into the RFP engine, Azure configuration follow-up, LLM enrichment, retrieval smoke testing, and clearer documentation of fallback and prototype behavior.

4.4 Week 4: Final Prototype Stabilization and Architecture-Complete RFP Flow

In Week 4, after CGI provided the remaining Azure configuration values, the team moved from a pol-

ished fallback prototype toward an architecture-complete stakeholder-facing prototype. The goal was still not a production-ready or business-validated assignment system, but rather to complete the intended prototype architecture: real capacity-aware RFP recommendations, Azure OpenAI embedding and Chroma retrieval in the active RFP flow when configured, Azure OpenAI chat enrichment for summary and explanation fields, preserved local fallback behavior, and clear limitations.

Yixiao led final integration QA, scope freeze, and documentation. This work defined the final prototype scope, demo path, claims the team can honestly make, claims the team should not overstate, architecture compliance notes, reproducibility notes, demo checklist, QA checklist, final limitations, and Week 4 report notes. The documentation distinguishes real local processing, heuristic scoring, Azure-backed retrieval, fallback retrieval, mock or sample data, Azure OpenAI chat enrichment, and remaining production-readiness limitations. Yixiao also coordinated final QA evidence from the other team members so the report could accurately reflect the implemented system rather

than overclaiming business validation.

Kian led final data validation and retrieval-evidence validation. Kian re-checked the proposal and response data structure, chunk metadata, and proposal-to-director / proposal-to-opportunity linkage limitations. For the architecture-complete Week 4 flow, Kian implemented a reproducible evidence workflow in `src/rfp_evidence_validator.py` and `scripts/generate_rfp_retrieval_evidence.py`, documented the results in `docs/rfp_final_evidence_validation.md`, and exported review artifacts under `outputs/rfp_evidence/`. The evidence run evaluated eight representative query themes (cloud migration, managed services, data analytics dashboard, cybersecurity compliance, application modernization, service desk support, change management training, and short vague requests), compared Azure / Chroma retrieval against local fallback on three sample themes, and recorded top retrieved chunk IDs, relevance notes, and weak or misleading retrieval cases using redacted previews rather than raw proposal text. Because full-corpus cloud embedding was constrained by service rate limits during the evidence run, the Azure / Chroma path was validated on a representative chunk subset, while local fallback searched the full historical corpus. A key limitation remains that retrieved proposal chunks should not be overinterpreted as proven director-experience evidence when `opportunity_owner` or `opportunity_id` linkage is incomplete in all validated chunks.

Lyken led the Azure / Chroma retrieval backend integration. After the required Azure configuration values became available, Lyken confirmed the Azure environment, validated Azure embedding access, and moved the Azure OpenAI embedding plus Chroma retrieval path from smoke-test status into the active RFP backend. The RFP engine now prefers Azure-generated embeddings and Chroma retrieval when the required configuration is available, while preserving deterministic local fallback retrieval if Azure embedding, Chroma storage, or Chroma querying is unavailable. Lyken also documented retrieval-mode behavior, vector-store rebuild/query steps, backend limitations, and the fallback path that keeps the final demo reproducible.

Freya led final RFP output quality, dashboard polish, and Azure OpenAI chat enrichment. The RFP page was updated to a Week 4 final prototype presentation, including clearer retrieval,

capacity-data, and recommendation-status labels; upload/paste workflow labels workflow labeling; a consolidated capacity/retrieval notice; and severity-specific risk flag styling. Freya connected the RFP page to real or pre-built capacity data by passing `capacity_df` into `generate_assignment_context()` via `director_df=capacity_df`. This allows recommended directors to use real capacity signals when available rather than relying only on mock director data. When Azure chat is configured and RFP text is non-empty, `generate_assignment_context()` can use the chat deployment to enrich the RFP summary, effort estimate, and per-director match reasons. If chat enrichment is unavailable, raises an error, returns malformed JSON, returns partial JSON, or returns invalid nested fields, the system falls back to deterministic heuristic output while preserving the same dashboard-facing `assignment_context` schema. The output also preserves backward compatibility by keeping `effort["rationale"]` aligned with `effort["reason"]`.

Jai led RFP page implementation support and stability testing. Jai verified that the Streamlit RFP page can display the agreed `assignment_context` fields, including RFP summary, Estimated Effort, retrieved examples, similar RFPs, recommended directors, risk flags, and notes. Jai also checked that recommended-director cards can display capacity label, capacity score, relative load, assignment score, match reason, capacity explanation, experience-match explanation, and supporting chunks. For the architecture-complete flow, Jai verified that the page labels the active retrieval mode, remains stable when Azure / Chroma falls back to local retrieval, handles empty input and missing fields, and does not crash when LLM enrichment or capacity data is unavailable.

A key Week 4 integration fix connected the Director Capacity Dashboard output to the RFP Assignment Tool. The Streamlit app was already loading `capacity_df`, but the RFP page had previously called `generate_assignment_context(rfp_text)` without passing the loaded capacity data into the assignment engine. The call was updated to `generate_assignment_context(rfp_text, director_df=capacity_df)`, allowing recommended directors to use real or pre-built capacity signals when available. The RFP results area also tracks and displays the capacity

data source used for the recommendation, distinguishing live CRM capacity data, pre-built `director_capacity_df.csv`, and synthetic mock or fallback capacity data.

After the Azure embedding and Chroma retrieval path was integrated, the active RFP flow became architecture-complete at prototype level. The preferred path now uses Azure OpenAI embeddings and Chroma retrieval when configured, then combines retrieved examples with real capacity data and optional Azure chat enrichment to produce the dashboard-facing `assignment_context`. The local fallback retrieval path remains available for reproducibility and failure handling. The Streamlit RFP page labels retrieval status so stakeholders can distinguish Azure-backed retrieval from fallback retrieval.

Local QA and manual demo validation were completed after the Week 4 changes. The Director Capacity Dashboard launched successfully, loaded live CRM capacity data, and displayed director-level capacity outputs including capacity labels, KPI cards, a relative-load chart, and a label-mix chart. The RFP Assignment Tool accepted pasted sample RFP text, retrieved relevant examples through the active retrieval path when configured, generated an RFP summary, displayed estimated effort, recommended directors using live or pre-built capacity data when available, displayed risk flags, and labelled the retrieval and capacity-data status. The final full-suite test result should be reported from the last Week 4 QA run.

Overall, Week 4 moved the project from a Week 3 fallback / heuristic RFP prototype to an architecture-complete stakeholder-facing capstone prototype. The capacity dashboard remained stable around the relative-load-first interpretation. The RFP Assignment Tool became capacity-aware when real or pre-built capacity data is available, integrated Azure OpenAI embedding and Chroma retrieval into the active RFP flow when configured, gained Azure OpenAI chat enrichment for summary, effort, and match-reason fields, preserved local fallback behavior for robustness, and added stronger empty-state, error-state, retrieval-quality, and response-quality checks.

4.5 Week 5: User-Style Testing, Evidence Review, and Final Refinement

In Week 5, the team shifted from architecture-complete feature development toward user-style testing, issue fixing, CGI feedback tracking, and

final refinement. The goal was not to add major new functionality, but to validate whether the stakeholder-facing prototype could be demonstrated clearly, whether the RFP outputs were supported by appropriate evidence, and whether the app wording avoided overclaiming production readiness, director experience, or assignment authority.

Yixiao led final app QA and scope tracking. This work consolidated manual QA results, automated test status, issue / fix / decision tracking, teammate follow-up, CGI feedback placeholders, and final scope / caveat alignment. The Week 5 automated baseline was recorded as 149 passed, 2 warnings. Manual QA confirmed that the Director Capacity Dashboard loaded successfully, displayed computed CRM-derived capacity data, rendered key charts, and used appropriate prototype framing. The RFP Assignment Tool was browser-tested with three pasted sample inputs covering cloud migration / managed services, cybersecurity / compliance, and a vague short technology request. This work also reconfirmed that the RFP path passes `director_df=capacity_df` into `generate_assignment_context()` when capacity data is available, while documenting remaining caveats around upload input, empty input, large input, CGI feedback, and full browser-based Azure/Chroma app-flow verification. Yixiao also implemented a low-risk performance and stability refinement for interactive RFP retrieval: the app no longer rebuilds the Chroma collection by default on every Analyze RFP request, attempts to reuse an existing Chroma store or collection when available, supports explicit rebuild behavior through a configuration flag, and falls back to local retrieval if Chroma is unavailable, not installed, missing, or fails during reuse. This keeps the basic Streamlit demo path robust without making chromadb a hard runtime dependency, while preserving Azure/Chroma backend validation when dependencies and configuration are available.

Kian led evidence and output review. The review tested three representative RFP inputs and checked whether retrieved examples supported the displayed outputs. The cloud migration/managed services case produced mixed evidence because some highly ranked chunks were semantically related but came from broader or differently titled proposal material. The cybersecurity/compliance case produced mostly relevant evidence. The vague short input produced weak or broad evidence, rein-

Track	Owner	Week 4 focus	Main output
RFP contract consistency pass	Yixiao, Freya	Tightened the Week 3 RFP assignment contract before final integration.	Stable assignment_context; effort.reason primary and rationale alias; safer fallback chunks; non-finite numeric cleanup; regression tests; 108 passed.
Final integration QA	Yixiao	Froze final prototype scope, demo path, safe claims, limitations, and reproducibility notes.	QA checklist, demo checklist, architecture-compliance notes, final caveats, and report-ready scope language.
RFP data validation	Kian	Validated proposal structure, chunk metadata, retrieval examples, and linkage limits.	Final validation summary, representative retrieval examples, chunk metadata notes, and proposal-to-director / opportunity linkage caveats.
Azure / Chroma retrieval	Lyken	Integrated Azure OpenAI embeddings and Chroma retrieval into the active RFP flow, with fallback preserved.	Azure environment check, embedding validation, Chroma retrieval path, retrieval-mode notes, and fallback behavior.
RFP output quality and LLM enrichment	Freya	Polished RFP UI, wired real capacity data, and added Azure chat enrichment.	Week 4 labels, upload/paste labels, severity risk flags, capacity_df handoff, LLM-enriched summary/effort/match reasons, and fallback-safe tests/docs.
RFP page stability	Jai	Verified page display and stability across normal, fallback, missing-field, empty, and error states.	Assignment-context display checks, sample RFP tests, retrieval-mode display, and stability notes.
Final prototype validation	Freya	Finalizes the architecture-complete prototype.	Improved upload, retrieval, service-fit, LLM enrichment, and fallback behavior; documented in final_prototype_validation.md.

Table 5: Week 4 progress toward final prototype stabilization and architecture-complete RFP assignment flow.

forcing the need for low-confidence warnings when the submitted RFP text lacks detail. A key Week 5 conclusion is that retrieved chunks are useful as semantic evidence, but they should not be treated as proof of prior director involvement, confirmed director experience, or CRM opportunity ownership because reliable proposal-to-director and proposal-to-opportunity linkage is still unavailable.

Lyken led retrieval and configuration validation. The Week 5 checks confirmed that the local Azure configuration is readable, Azure embedding calls return valid vectors, Azure/Chroma retrieval works on both sample and full proposal-corpus backend checks, and generate_assignment_context() can return retrieval_mode and retrieval_status metadata. The full-corpus backend validation used 1,216 proposal chunks and returned retrieval_mode = azure_chroma. The scripted Azure/Chroma smoke test also passed. The local fallback path remains available and is still clearly distinguished from Azure/Chroma retrieval. Week

5 documentation therefore treats Azure/Chroma as validated at backend smoke-test, sample, and full-corpus levels, while avoiding a stronger claim of full browser-based Streamlit Azure/Chroma app-flow validation unless that exact UI path is explicitly run and documented.

Freya led stakeholder-facing usability and wording refinement. The capacity dashboard screenshots were reviewed as understandable and reasonable for stakeholder demonstration. For the RFP Assignment Tool, Freya refined wording and display choices to reduce overclaiming and improve interpretability. The results section label was changed from Similar Historical RFPs to Retrieved Supporting Examples, retrieved-result scores were described as match scores, fallback captions were rewritten in client-facing language, and the corpus caption was generalized from pasted-RFP query chunks to submitted RFP query text with historical or sample corpus chunks. Freya also separated retrieval status from LLM enrichment status so that local retrieval fallback is not confused with Azure

OpenAI chat fallback. Recommended-director metrics were rounded, and LLM match-reason guidance was updated so that director rationales are framed as prototype fit based on capacity, service-fit signals, and retrieved semantic evidence rather than proven prior experience.

Jai led assignment logic review and RFP status documentation. Jai tested recommendation behavior with varied RFP inputs, including cloud/data modernization, cybersecurity, vague IT consulting, procurement/deadline-only text, a large multi-domain transformation RFP, empty input, TXT upload, and fallback mode. These tests confirmed that the tool can produce coherent outputs across several realistic and stress-test scenarios, while also documenting remaining caveats. First-run real-corpus Azure/Chroma indexing can be slow because the app may rebuild the full proposal index during analysis, so caching or prebuilding the Chroma index remains a recommended follow-up. Vague or broad RFP text can still produce confident-looking service-fit signals, so stronger low-detail risk flags may be useful. Jai also checked proposal-to-CRM crosswalk candidates and found no strong automatic CRM match for most proposal/RFP entries, supporting the current caveat that retrieved chunks should be treated as semantic evidence rather than confirmed director-history evidence.

Overall, Week 5 refined the final prototype through a validation and de-risking loop: retrieval/config checks confirmed that the preferred backend path can operate with fallback safety; evidence review clarified which outputs are supported by retrieved examples; assignment wording was softened where needed; stakeholder-facing labels and captions were made clearer; and final QA tracking consolidated test results, open caveats, and safe report language. The resulting system is better positioned for CGI review because it distinguishes capacity-aware recommendation signals, retrieved semantic evidence, LLM-enriched wording, local fallback behavior, and remaining business-validation limits.

5 Current Methodology and Prototype Design

The current capacity model is transparent and formula-based. Current workload is represented as the sum of sales load and inferred delivery load. Sales load uses open opportunities, stage weight, probability, authoritative revenue, and project dura-

tion. Delivery load uses won active deliveries and monthly revenue as a proxy for delivery commitment. This approach keeps the scoring logic interpretable and avoids using an LLM as the source of truth for numeric capacity scores.

In the Week 5 refined prototype, `relative_load` remains the primary business-facing capacity metric. It is defined as a director's current workload divided by that same director's historical average workload. This interpretation is useful for communicating overextension because it can distinguish a director who is slightly above normal workload from one who is carrying multiple times their historical norm. `capacity_score` remains in the backend output for compatibility and for downstream RFP assignment scoring, but it is no longer treated as the main dashboard metric because it is capped once `relative_load` reaches or exceeds the baseline.

Capacity labels are assigned from `relative_load`. The current prototype labels directors as *Available*, *Near Historical Norm*, *High Load*, *Overextended*, or *No baseline*. These thresholds are project-side calibration choices based on the observed director-level distribution and the business interpretation that carrying at least twice the historical workload should be considered overextended. CGI should review and adjust these thresholds before production use.

The dashboard filtering logic is designed to keep the historical baseline stable. Territory and opportunity-owner filters define the selected director population for both current workload and historical baseline. Row-level filters such as status, sales stage, opportunity type, sales model, and created-date range narrow the current workload and trend views, but do not recompute the historical baseline denominator. This allows filtered views to answer how a selected slice of current workload compares with the selected directors' normal workload.

The Streamlit dashboard delegates capacity calculations to shared functions in `capacity_engine.py`. For live filtered views, the interface rebuilds current workload from filtered opportunity rows while keeping the historical baseline anchored to the selected owner / territory population. This keeps the scoring logic centralized in the engine while allowing the dashboard to support interactive filtering. By Week 5, the resulting `capacity_df` is passed into the RFP assignment engine when available, connecting the Director Capacity Dashboard output to

Track	Owner	Week 5 focus	Main output
Final app QA and scope tracking	Yixiao	Coordinated user-style QA, issue/fix/decision tracking, teammate follow-up, and final claim-boundary alignment.	Final QA tracker, automated baseline of 149 passed, 2 warnings, manual dashboard and RFP checks, scope/caveat notes, Chroma reuse and fallback-safe retrieval stability fix, and report-ready safe wording.
Evidence and output review	Kian	Reviewed whether retrieved examples support displayed RFP outputs across cloud, cybersecurity, and vague short input cases.	Evidence review notes, effort-support checks, weak-evidence cases, and semantic-evidence boundary language.
Retrieval and config validation	Lyken	Validated Azure configuration, Azure embedding, Azure/Chroma retrieval, retrieval metadata, and fallback behavior.	Backend smoke-test, sample, and full-corpus Azure/Chroma validation; retrieval_mode/retrieval_status reporting; fallback-safe retrieval behavior.
Dashboard usability and wording	Freya	Reviewed stakeholder-facing dashboard and RFP output clarity, including retrieval, fallback, LLM, and linkage caveat wording.	Retrieved Supporting Examples label, match-score wording, separate LLM enrichment status, rounded director metrics, and safer director-experience caveats.
Assignment logic review	Jai	Tested recommendation behavior, risk flags, fallback mode, upload behavior, and proposal-to-CRM linkage assumptions.	Varied RFP test notes, softened match-reason wording, fallback test result, remaining broad/vague-input caveats, and proposal-to-CRM crosswalk limitation notes.

Table 6: Week 5 progress from architecture-complete prototype development toward user-style testing, evidence review, final wording refinement, and scope/caveat consolidation.

capacity-aware director recommendations.

For RFP assignment, the current prototype combines proposal text preprocessing, retrieval-oriented evidence, structured capacity signals, service_solution fit, heuristic scoring, and optional LLM enrichment. The RFP pipeline extracts proposal or response text from `proposals_responses.json` when available, creates chunks for the local proposal corpus, and uses a built-in sample corpus only when the local proposal corpus is unavailable. New RFP input is accepted through pasted text or TXT, DOCX, and text-based PDF upload. The submitted RFP text is used as query text against the chunked historical or sample corpus, and the dashboard-facing `assignment_context` is produced from the retrieved supporting examples, capacity data, service-fit signal, risk flags, and optional LLM enrichment.

The intended RFP recommendation logic combines retrieval evidence with structured director capacity signals. In Week 4, the Streamlit RFP page was updated to call `generate_assignment_context(rfp_text, director_df=capacity_df)` when capacity data is available. This allows recommended direc-

tors to use real or pre-built capacity signals, including `capacity_label`, `capacity_score`, and `relative_load`, rather than relying only on mock director data. If only synthetic mock capacity data is available, the dashboard labels this clearly and the recommendation should be interpreted as fallback prototype guidance. Week 5 QA reconfirmed this capacity handoff and clarified that capacity-aware recommendations are still decision-support outputs requiring stakeholder validation rather than final assignment decisions.

The RFP output uses a stable `assignment_context` schema for dashboard display. This schema includes an RFP summary, an effort object displayed as Estimated Effort, retrieved supporting examples, recommended directors, risk flags, and notes. Recommended-director cards can display capacity label, capacity score, relative load, assignment score, match reason, capacity explanation, experience match explanation, and supporting chunks. This design keeps the RFP page stable even when individual backend components use real data, fallback data, heuristic logic, or optional LLM enrichment.

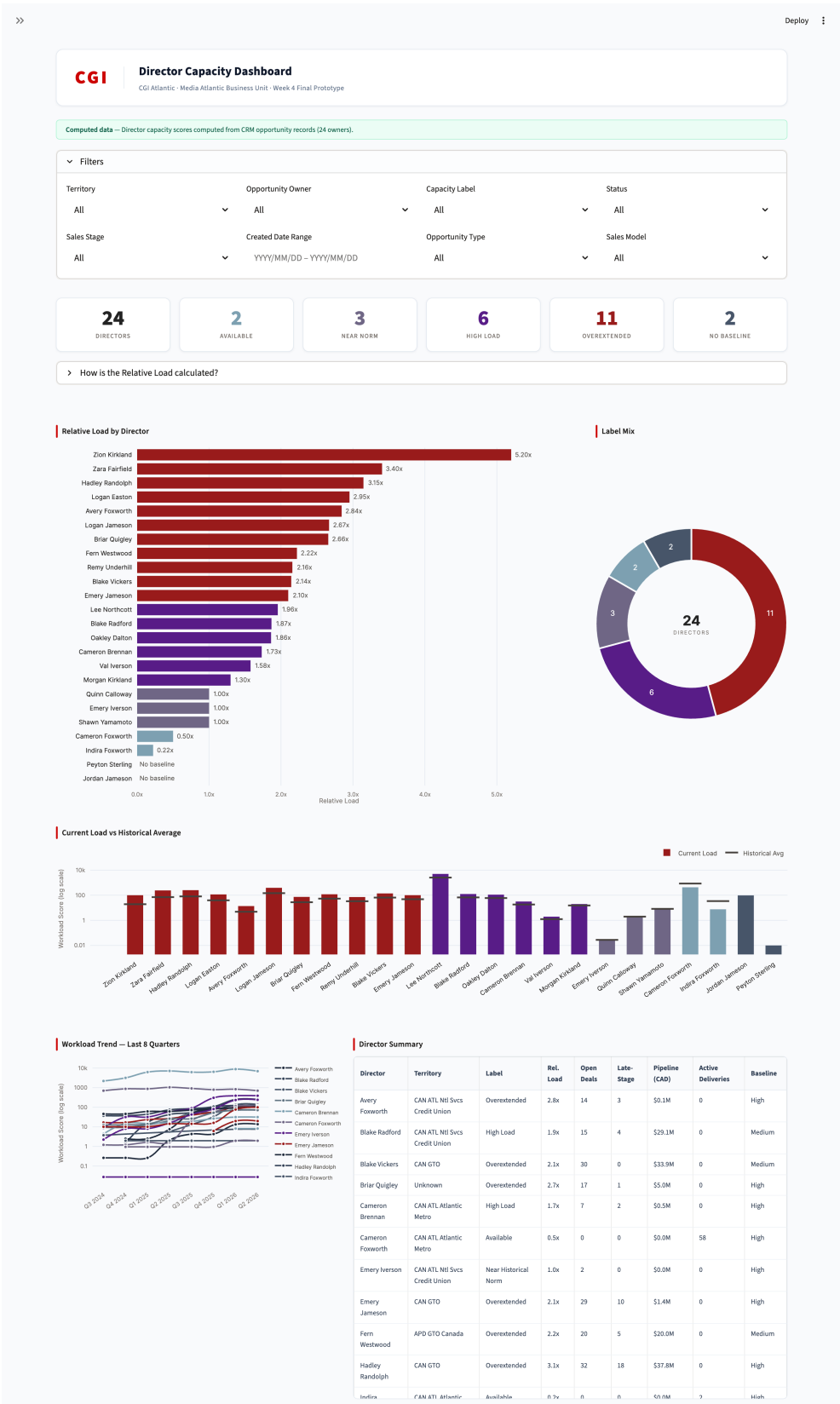
Week 4 also added Azure OpenAI chat enrich-

ment. When Azure chat configuration is available, `generate_assignment_context()` can use the chat deployment to enrich the RFP summary, effort estimate, and per-director match reasons. When chat enrichment is unavailable, fails, or returns malformed JSON, the system falls back to deterministic heuristic output using the same schema. This keeps the dashboard output robust while improving readability when LLM enrichment succeeds. LLM-generated fields are still treated as prototype-quality explanations rather than validated business analysis. Week 5 usability review also separated retrieval mode from LLM enrichment status in the RFP page so stakeholders can distinguish local retrieval fallback from Azure OpenAI chat fallback.

The preferred retrieval architecture uses Azure OpenAI embeddings with Chroma vector-store retrieval. After CGI provided the remaining Azure configuration values, Azure embedding access passed a real smoke test, sample Azure + Chroma retrieval was validated, and the Azure / Chroma retrieval adapter was integrated into the active RFP engine. Week 5 retrieval and configuration validation further confirmed that Azure/Chroma retrieval works at backend smoke-test, sample, and full-corpus levels, including retrieval over 1,216 proposal chunks and `retrieval_mode / retrieval_status` metadata returned through `generate_assignment_context()`. When Azure configuration is available and the preferred path succeeds, the RFP engine uses Azure-generated embeddings and Chroma retrieval. When Azure embedding, Chroma storage, Chroma querying, or vector-store setup is unavailable, the system falls back to deterministic local retrieval while preserving the same dashboard-facing `assignment_context` schema. The Streamlit RFP page reads the retrieval mode from the backend output, but full browser-based Azure/Chroma app-flow validation should only be claimed when that exact UI path has been explicitly run and documented.

Week 5 evidence review also refined the interpretation of retrieved examples. Retrieved proposal or response chunks can support semantic similarity discussion, but they do not currently prove prior director involvement, confirmed director experience, or CRM opportunity ownership because reliable proposal-to-director and proposal-to-opportunity linkage is still unavailable. The RFP page and report wording therefore describe these items as retrieved supporting examples rather than proven

historical director experience.



> How is the Relative Load calculated?

Relative Load by Director

Zion Kirkland

3.40x

Zara Fairfield

3.15x

Hadley Randolph

2.95x

Logan Easton

2.84x

Avery Foxworth

2.67x

Logan Jameson

2.66x

Briar Quigley

2.22x

Fern Westwood

2.18x

Renny Underhill

2.14x

Blake Vickers

2.10x

Emery Jameson

1.96x

Lee Northcott

1.87x

Blake Radford

1.86x

Oakley Dalton

1.73x

Cameron Brennan

1.58x

Val Iverson

1.30x

Morgan Kirkland

1.00x

Quinn Calloway

1.00x

Emery Iverson

1.00x

Shawn Yamamoto

0.50x

Cameron Foxworth

0.22x

Indira Foxworth

No baseline

Peyton Sterling

No baseline

Jordan Jameson

No baseline

Label Mix

24

DIRECTORS

Current Load vs Historical Average

Current Load

Historical Avg

Workload Trend — Last 8 Quarters

Avery Foxworth

Blake Radford

Blake Vickers

Briar Quigley

Cameron Brennan

Cameron Foxworth

Emery Iverson

Emery Jameson

Fern Westwood

Hadley Randolph

Indira Foxworth

Director Summary

Director	Territory	Label	Rel. Load	Open Deals	Late-Stage	Pipeline (CAD)	Active Deliveries	Baseline
Avery Foxworth	CAN ATL Ntl Svcs Credit Union	Overextended	2.8x	14	3	\$0.1M	0	High
Blake Radford	CAN ATL Ntl Svcs Credit Union	High Load	1.9x	15	4	\$29.1M	0	Medium
Blake Vickers	CAN GTO	Overextended	2.1x	30	0	\$33.9M	0	Medium
Briar Quigley	Unknown	Overextended	2.7x	17	1	\$5.0M	0	High
Cameron Brennan	CAN ATL Atlantic Metro	High Load	1.7x	7	2	\$0.5M	0	High
Cameron Foxworth	CAN ATL Atlantic Metro	Available	0.5x	0	0	\$0.0M	58	High
Emery Iverson	CAN ATL Ntl Svcs Credit Union	Near Historical Norm	1.0x	2	0	\$0.0M	0	High
Emery Jameson	CAN GTO	Overextended	2.1x	29	10	\$1.4M	0	High
Fern Westwood	APD GTO Canada	Overextended	2.2x	20	5	\$20.0M	0	Medium
Hadley Randolph	CAN GTO	Overextended	3.1x	32	18	\$37.8M	0	High
Indira	CAN ATL Atlantic	Available	0.7x	0	0	\$0.0M	7	High

Figure 1: Week 5 Director Capacity Dashboard prototype.

»

Deploy

CGI

RFP Assignment Tool

CGI Atlantic - Media Atlantic Business Unit - Week 4 Final Prototype

Computed data — Director capacity scores computed from CRM opportunity records (24 owners).

Week 4 Final Prototype — RFP Assignment Tool

Retrieval: active mode is shown after analysis (Azure/Chroma when available, local fallback otherwise). Capacity data: real director capacity scores when available. Recommendations: prototype guidance only — not final CGI assignment decisions.

RFP Input

Upload RFP document

20-09aa-final.pdf

96,148

+

Loaded text from 20-09aa-rfp-for-case-management-software-final.pdf.

Upload a supported document or paste RFP text directly below.

submitted by

e-mail to the Purchasing Office at purchasing@duluthmn.gov no

later than the date

indicated on the Calendar of Events. Answers to the questions will

be posted as an

Addendum to the RFP.

I-S. Addenda to the RFP. If the City deems it necessary to revise any

part of this RFP

before the successful response date, the City will make addenda

Territory

Service Domain

Any

Any

Territory and service domain are UI context only for now; backend scoring does not use them yet.

Analyze RFP

Retrieval mode — Azure/Chroma retrieval active.

Prototype output — capacity-aware recommendations using Computed from CRM opportunity records. Retrieval: Azure/Chroma retrieval active.

Azure/Chroma retrieval is active for this result.

Retrieval corpus: local proposals_responses.json corpus; 1216 corpus chunks; 11 submitted RFP query text; historical/sample corpus chunks.

Retrieved chunks cannot currently be tied reliably to a specific director or opportunity. Treat them as semantic similarity evidence, not proof of prior director experience.

LLM enrichment: Azure OpenAI chat was used for the summary, effort estimate, and director match wording.

RFP Summary

The City of Duluth is seeking proposals for a case management software solution for its City Attorney's Office, including software, hardware, implementation, training, and ongoing support to replace the current system and improve operational efficiency.

Estimated Effort

High

Estimated duration: 12-16 weeks

The project involves multiple components, including software and hardware acquisition, implementation, training, and long-term support, requiring significant coordination and expertise.

Retrieved Supporting Examples

1. 4_courts_case_management_system — match score 0.61

Proposal: 4_courts_case_management_system

Chunk ID: 4_courts_case_management_system_proposal_chunk_0031

Match score: 0.61

=100%). A Proponent who bids 159.00 receives 88% of the possible points for that category (120/150 = 80%), and a Proponent who bids 240.00 receives 50% of the possible points for that category (120/240 = 50%). Lowest rate ----- x Total available points = Score for second-lowest rate Second-lowest rate...

2. 3_data_sharing_platform_shared_analytics — match score 0.59

3. 4_courts_case_management_system — match score 0.59

Recommended Directors

1. Indra Foxworth

Capacity: Available

Capacity score: 0.78

Relative load: 0.22x

Assignment score: 0.71

Prototype fit is based on available capacity, service-fit signals, and retrieved semantic evidence; it does not prove prior director experience. LLM note: Indra's background in managing complex, multi-phase projects makes her a strong candidate to oversee the implementation and integration of the proposed case management solution.

Capacity explanation

Indra Foxworth is labelled Available with capacity score 0.78 and relative load 0.22. This is a heuristic input to the assignment score, not a final staffing decision.

Experience match explanation

Experience fit is estimated from retrieved historical/sample RFP chunks and simple service-keyword overlap (0.40).

Supporting chunks: 4_courts_case_management_system_proposal_chunk_0031, 3_data_sharing_platform_shared_analytics_proposal_chunk_0010

2. Cameron Foxworth

3. Jordan Jameson

Risk Flags

Low: Retrieved chunks cannot currently be tied reliably to a specific director or opportunity. Treat them as semantic similarity evidence, not proof of prior director experience.

Medium: Some directors have no historical baseline, so capacity score and relative load use heuristic defaults in RFP ranking. Review No baseline candidates before assignment.

High: At least one director in the broader capacity pool is overextended based on capacity signals. Review workload before assigning outside the recommended shortlist.

Notes

Prototype output enriched with Azure OpenAI chat analysis. Retrieval used Azure/Chroma. Retrieved evidence searched the local proposals_responses.json corpus.

Figure 2: Week 5 RFP Assignment Tool prototype.

6 Project Management and Next Steps

In Week 5, the team's project management focus shifted from architecture-complete prototype development to final app QA, user-style testing, issue fixing, CGI feedback tracking, and final refinement. The team treated the Director Capacity Dashboard and RFP Assignment Tool as stakeholder-facing prototype deliverables and focused on whether the app could be demonstrated clearly, whether outputs were supported by appropriate evidence, and whether the wording avoided overclaiming production readiness, director experience, Azure-backed behavior, or assignment authority.

Yixiao led final app QA and scope tracking. This work consolidated manual QA results, automated test status, issue / fix / decision tracking, teammate follow-up, CGI feedback placeholders, and final scope / caveat alignment. The Week 5 automated baseline was recorded as 149 passed, 2 warnings. Manual QA confirmed that the Director Capacity Dashboard loaded successfully, displayed computed CRM-derived capacity data, rendered key charts, and used appropriate prototype framing. The RFP Assignment Tool was browser-tested with three pasted sample inputs covering cloud migration / managed services, cybersecurity / compliance, and a vague short technology request. This work also reconfirmed that the RFP path passes `director_df=capacity_df` into `generate_assignment_context()` when capacity data is available, while documenting remaining caveats around upload input, empty input, large input, CGI feedback, and full browser-based Azure/Chroma app-flow verification.

Kian led Week 5 evidence and output review. Kian reviewed three representative RFP inputs and checked whether retrieved examples supported the displayed summaries, effort estimates, match reasons, and risk flags. The review found that cloud migration / managed services produced mixed but usable semantic evidence, cybersecurity / compliance produced mostly relevant evidence, and vague short input produced broad or weak evidence that should be treated as low confidence. Kian also clarified that effort estimates remain heuristic: retrieved examples only increase effort when the best similarity is below the novelty threshold, so retrieved examples should not be described as validating delivery duration. The key documentation outcome was a clearer evidence boundary: retrieved chunks are semantic evidence related to the submitted RFP

text, not proof of prior director involvement, confirmed director experience, or CRM opportunity ownership.

Lyken led retrieval and configuration validation. Week 5 checks confirmed that Azure configuration is readable locally, Azure embedding calls return valid vectors, Azure/Chroma retrieval works on sample and full proposal-corpus backend checks, and `generate_assignment_context()` can return `retrieval_mode` and `retrieval_status` metadata. The full-corpus backend validation used 1,216 proposal chunks and returned `retrieval_mode = azure_chroma`. The scripted Azure/Chroma smoke test also passed, while the deterministic local fallback path remained available and clearly distinguished from Azure/Chroma retrieval. The main project-management caveat is that these checks validate the backend retrieval path at smoke-test, sample, and full-corpus levels; full browser-based Streamlit Azure/Chroma app-flow validation should only be claimed if that exact UI path is explicitly run and documented.

Freya led stakeholder-facing usability and wording refinement. Freya reviewed the final dashboard and RFP page screenshots from the perspective of a CGI stakeholder and confirmed that the Director Capacity Dashboard was understandable for stakeholder demonstration. For the RFP page, Freya identified and implemented wording and display refinements that reduce overclaiming and improve interpretability. These refinements included changing Similar Historical RFPs to Retrieved Supporting Examples, changing similarity score to match score, rewriting fallback captions in client-facing language, generalizing pasted-RFP query wording to submitted RFP query text, separating retrieval mode from LLM enrichment status, rounding recommended-director metrics, and updating LLM match-reason guidance so director rationales are framed as prototype fit based on capacity, service-fit signals, and retrieved semantic evidence rather than proven prior experience.

Jai led assignment logic review and RFP status documentation. Jai tested recommendation behavior across varied RFP inputs, including cloud / data modernization, cybersecurity, vague IT consulting, procurement / deadline-only text, a large multi-domain transformation RFP, empty input, TXT upload, and fallback mode. These tests confirmed that the RFP Assignment Tool can produce coherent outputs across several realistic and stress-test scenarios while preserving caveats. Jai also docu-

mented remaining implementation and interpretation risks: first-run real-corpus Azure/Chroma indexing can be slow because the app may rebuild the proposal index during analysis; vague or broad RFP text can still produce confident-looking service-fit signals; broad multi-domain inputs may retrieve several chunks from one proposal family; and fallback similarity scores are not directly comparable to Azure/Chroma scores. A local proposal-to-CRM crosswalk check found no strong automatic CRM match for most proposal/RFP entries, supporting the current caveat that retrieved chunks should be treated as semantic evidence rather than confirmed director-history evidence unless CGI provides or validates a reliable mapping.

The next stage of the project should focus on final stakeholder review, demo polish, handoff documentation, and validation rather than major new feature expansion. The capacity dashboard thresholds and label names should be reviewed with CGI to confirm whether they match business language. The RFP Assignment Tool should be evaluated with representative stakeholder examples, with particular attention to whether retrieved supporting examples are relevant, whether risk flags are useful, whether vague-input warnings are strong enough, and whether recommendation wording remains appropriately caveated. Future work should further production-harden the Azure + Chroma retrieval path, including more robust index persistence, cache management, and deployment-time dependency handling; expand retrieval validation across more representative RFPs; connect reliable proposal-to-CRM or proposal-to-director metadata if CGI can provide it; improve document upload parsing for supported file types; and calibrate recommendation weights with CGI feedback.

7 Current Assumptions and Risks

Several assumptions and risks remain at the end of Week 5. First, `opportunity_id` is still treated as the required join key for opportunity-level merging. Second, `opps2` remains the primary base CRM opportunity table, while `opps1` is used only for selected supplemental fields after being collapsed to one row per `opportunity_id`. Third, `opportunity_owner` is treated as the main director-level entity for MVP capacity scoring, while `opportunity_manager` is retained as optional metadata.

The capacity dashboard still depends on in-

ferred workload rather than measured hours, calendar commitments, or manually confirmed director capacity. Current load is estimated from sales pipeline and inferred delivery signals, so the resulting workload estimates should be treated as directional decision-support indicators rather than exact measurements of utilization, workload hours, or guaranteed availability.

The Week 5 refined dashboard continues to use `relative_load` as the primary business-facing workload metric. This improves interpretability, but the current label thresholds remain temporary project-side calibration choices. Labels such as Available, Near Historical Norm, High Load, Overextended, and No baseline should be reviewed by CGI before production use. In particular, the threshold for Overextended reflects the current team interpretation that at least twice the historical workload should be treated as overextension.

The stable-baseline filtering decision also remains an assumption. Territory and opportunity-owner filters define the selected director population for both current workload and baseline, while row-level filters narrow current workload without recomputing the historical baseline. This makes the dashboard easier to interpret, but CGI may later decide that some additional filters should also redefine the baseline population.

The RFP Assignment Tool is capacity-aware when the Streamlit page passes `capacity_df` into `generate_assignment_context()`. Week 5 QA reconfirmed this app-facing capacity handoff. However, the resulting recommendations are still prototype decision-support guidance rather than final CGI staffing decisions. The assignment score combines available capacity signals, retrieval evidence, keyword or service-domain overlap, and heuristic logic. These weights have not yet been calibrated against stakeholder-approved outcomes or historical assignment labels.

The project still lacks ground-truth labels for director capacity, RFP effort, and correct director assignment. Evaluation therefore relies on stakeholder review, threshold sensitivity checks, sanity checks, smoke tests, manual demo validation, evidence review, and selected case studies rather than standard supervised-learning metrics.

The RFP retrieval evidence has important limitations. Proposal chunks cannot currently be reliably linked to directors or opportunities when `opportunity_owner` and `opportunity_id` are missing from validated chunk metadata. Week

5 evidence review and proposal-to-CRM cross-walk checks reinforced this limitation: most proposal/RFP entries do not have strong automatic CRM matches, and weak theme-based candidates are not reliable enough to use as confirmed director-to-RFP evidence. As a result, retrieved examples can support semantic similarity claims, but they should not be overinterpreted as proven director-experience evidence, confirmed director involvement, or CRM opportunity ownership.

Retrieval quality also depends on the specificity of the submitted RFP text. Week 5 evidence review found that cybersecurity / compliance inputs produced mostly relevant retrieved examples, while cloud migration / managed services produced mixed evidence and vague short inputs produced broad or weak evidence. Short or underspecified inputs can still generate confident-looking summaries, match scores, or service-fit signals. The RFP page should therefore continue to show low-confidence warnings for vague input and should encourage stakeholders to provide enough scope detail before relying on recommendations.

Azure embedding access and Azure + Chroma retrieval were validated after CGI provided the remaining Azure configuration values, and Week 5 retrieval/configuration checks confirmed that the backend path works at smoke-test, sample, and full-corpus levels. The full-corpus backend validation used 1,216 proposal chunks and returned `retrieval_mode = azure_chroma`. However, this integration remains prototype-level rather than production-hardened. The report should avoid claiming a fully browser-validated Azure/Chroma Streamlit app flow unless that exact UI path has been explicitly run and documented. The system preserves local fallback retrieval when Azure embedding, Chroma storage, Chroma querying, or vector-store setup is unavailable.

Real-corpus Azure/Chroma retrieval still has a performance and dependency-management caveat, but Week 5 added a low-risk stability fix for the interactive path. The app no longer rebuilds the Chroma collection by default on every RFP analysis request. Instead, the retrieval code attempts to reuse an existing Chroma store or collection when available, supports explicit rebuild behavior through a configuration flag, and falls back to deterministic local retrieval when Chroma is unavailable, not installed, missing, or fails during reuse. This makes the basic Streamlit demo path more robust and avoids making chromadb a hard runtime depen-

dency. First-time index creation, explicit rebuilds, or production-scale index management may still require further optimization.

Azure OpenAI chat enrichment introduces an additional interpretation risk. When configured, the LLM can enrich the RFP summary, effort estimate, and director match reasons. These fields improve readability, but they are not validated business analysis. If chat enrichment fails or returns malformed output, the system falls back to deterministic heuristic output. Week 5 usability review separated retrieval mode from LLM enrichment status so stakeholders do not confuse local retrieval fallback with Azure OpenAI chat fallback. The dashboard and report should continue to label LLM-enriched text as prototype explanation rather than authoritative staffing guidance.

The RFP page still has interface limitations. Week 5 testing confirmed that pasted input and TXT upload can support stakeholder-style testing, and the RFP upload/paste workflow is understandable from a user perspective. However, production-grade document ingestion remains future work. DOCX and text-based PDF parsing should continue to be tested carefully, scanned PDFs may require manual paste or OCR outside the app, and large or malformed inputs require additional validation. Upload support should therefore be described conservatively as prototype-level rather than production-grade document processing.

Finally, the system may combine real data, heuristic logic, fallback retrieval, sample data, mock capacity data, Azure smoke-test components, Azure/Chroma backend retrieval, and LLM-enriched text during development and demonstration. These components must remain clearly labeled in the dashboard and documentation so that stakeholders understand which parts are implemented, which parts are fallback or heuristic, which parts are validated only at backend or sample level, and which parts still require future integration or business review.

8 Conclusion

This Week 5 version extends the report from a Week 4 architecture-complete prototype update into a refined final stakeholder-facing prototype report. Week 1 established the reproducible opportunity merge, preserved opportunity-level grain, and created the main module skeletons. Week 2 refined the cleaned opportunity data, added

validation-oriented field assessment, produced an initial director-level capacity scoring output, connected the dashboard to capacity results, and advanced the RFP pipeline toward preprocessing, chunking, and retrieval-style output. Week 3 refined the Director Capacity Dashboard around a `relative_load`-first interpretation and shifted the main development focus toward RFP assignment, including fallback retrieval, assignment schema stabilization, heuristic recommendation logic, and Streamlit RFP page integration. Week 4 stabilized the prototype into an architecture-complete stakeholder-facing version with capacity-aware RFP recommendations, Azure OpenAI chat enrichment, Azure OpenAI embedding and Chroma retrieval integration, preserved local fallback behavior, upload/paste input support, and clearer prototype-status labeling.

In Week 5, the team focused on final app QA, user-style testing, evidence review, assignment caveat refinement, stakeholder-facing wording polish, and final scope tracking rather than major new feature development. The capacity dashboard remained centered on transparent formula-based workload estimation and relative-load interpretation. The RFP Assignment Tool was further validated and refined: the app-facing path was reconfirmed to pass real or pre-built `capacity_df` into `generate_assignment_context()` when available, Azure/Chroma retrieval was validated at back-end smoke-test, sample, and full-corpus levels, local fallback retrieval remained available, and the RFP page was updated to distinguish retrieval mode from LLM enrichment status. Week 5 evidence review also clarified that retrieved supporting examples should be treated as semantic evidence related to the submitted RFP text, not as proof of prior director involvement, confirmed director experience, or CRM opportunity ownership.

Overall, the project now demonstrates a transparent decision-support workflow: a stakeholder can view director capacity results, inspect workload drivers, submit or paste RFP text, receive capacity-aware prototype recommendations when capacity data is available, review retrieved supporting examples, and see risk flags and caveats. The system remains an applied stakeholder-facing prototype rather than a production-ready or business-validated assignment system. Future work should focus on CGI review of capacity thresholds and label language; calibration of recommendation weights; larger-scale validation of retrieved exam-

ples; reliable proposal-to-director or proposal-to-CRM linkage; further production hardening of the Azure + Chroma retrieval path, including more robust index persistence, cache management, and deployment-time dependency handling; continued document upload testing; stronger low-detail input warnings; and final handoff documentation.